Truncation Errors and the Taylor Series (Chapter 4)

The Taylor Series

- Provides a means of predicting a function value at one point in terms of the function value and its derivatives at another point.
- The Thm states that any smooth function can be approximated as a polynomial.

Taylor's Thm:

$$f(x) = f(a) + f'(a)(x-a) + \frac{1}{2!}f''(a)(x-a)^2 + \frac{1}{3!}f'''(a)(x-a)^3 + \frac{1}{n!}f^{(n)}(a)(x-a)^n + R_n$$

($f(x)$: x_{i+1}; $f(a)$: x_i)

$$R_n = \int_a^x \frac{(x-t)^n}{n!} f^{(n+1)}(t)\,dt$$, t is a dummy variable.

(if and its first $(n+1)$ derivatives are continuous on an interval containing a and x.)

Term by term

First term $f(x_{i+1}) \cong f(x_i)$ → zero-order approximation (constant)

$f(x_{i+1}) \cong f(x_i) + f'(x_i)(x_{i+1} - x_i)$

→ first order approximation (straight line)

$$f(x_{i+1}) \cong f(x_i) + f'(x_i)\underbrace{(x_{i+1}-x_i)}_{h} + \underbrace{\frac{f''(x_i)}{2}(x_{i+1}-x_i)^2}_{\text{second-order term}}$$

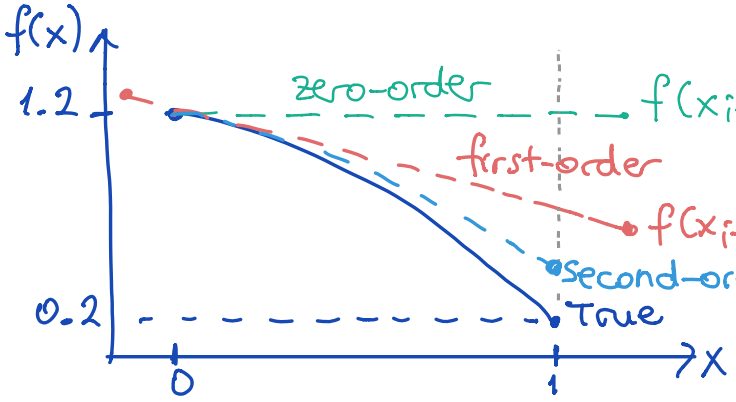
$$+ \frac{f^{(n)}(x_i)}{n!}(x_{i+1}-x_i)^n + R_n \qquad h = x_{i+1}-x_i \text{ (step size)}$$

$$R_n = \frac{f^{(n+1)}(\xi)}{(n+1)!}\underbrace{(x_{i+1}-x_i)}_{h}^{n+1} = O(h^{n+1})$$

Ex $f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$

Use zero to 4th order taylor series expansion to approximate $f(x)$.

$x_i = 0$ with $h = 1$. That is, predict $f(x)$ at $x_{i+1} = 1$



$f(x_{i+1}) \cong f(x_i) = f(0)$

$f(x_{i+1}) \cong f(x_i) + f'(x_i)h$

$f(x_{i+1}) \cong f(x_i) + f'(x_i)h + f''(x_i)\frac{h^2}{2!}$

$f(0) = 1.2$

$f(1) = 0.2 \rightarrow$ true value we're trying to predict.

order	Approximation	Truncation Error at x=1
n=0	$f(1) \cong 1.2$	$E_t = 0.2 - 1.2 = -1.0$

n=1

$$f'(x)\Big|_{x=0} = -(0.4)x^3 - 0.45x^2 - 1x - 0.25\Big|_{x=0}$$

$f'(0) = -0.25$

$f(x_{i+1}) \cong 1.2 - 0.25h$

$f(1) = 0.95 \longrightarrow E_t = 0.2 - 0.95 = -0.75$

n=2

$f''(0) = -1.0$

$f(x_{i+1}) \cong 1.2 - 0.25h - 0.5h^2$

$f(1) = 0.45 \rightarrow E_t = 0.2 - 0.45 = -0.25$

* $R_n = O(h^{n+1})$ → proportional to h^{n+1}

e.g. if error is $O(h)$, halving the step size will halve the error

or $O(h^2)$ → halving h will quarter the error.

Numerical Differentiation

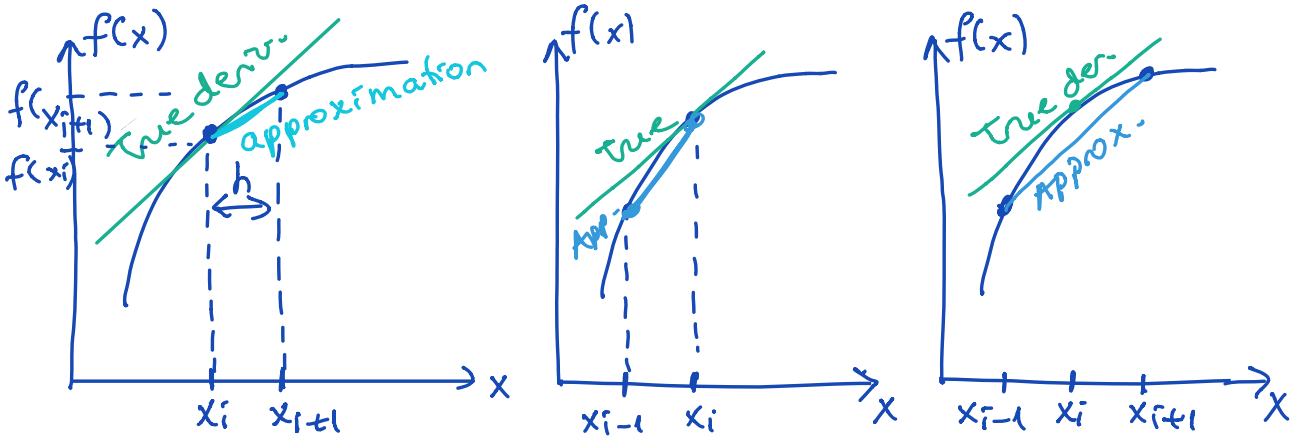
Finite Divided Difference:

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} + O(\overbrace{x_{i+1} - x_i}^{h})$$

$$f'(x_i) = \frac{\Delta f_i}{h} + O(h)$$

Δf_i → first forward difference

h → step size



Forward finite-divided difference | Backward finite-divided diff. | Centered finite divided-diff.

Backward Difference Approximation of the First Derivative:

$$f(x_{i-1}) = f(x_i) - f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \cdots \quad (1)$$

$x_{i-1} - x_i = -h$

$$f'(x_i) \cong \frac{f(x_i) - f(x_{i-1})}{h}$$ Error is $O(h)$

Centered Difference Approximation of the First Derivative

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \cdots \quad (2)$$

Subtract (1) from (2):

$$f(x_{i+1}) - f(x_{i-1}) = 2f'(x_i)h + \frac{2f^{(3)}(x_i)}{3!}h^3 + \cdots$$

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h} - \underbrace{\frac{f^{(3)}(x_i)}{6}h^2 + \cdots}_{-O(h^2)}$$

↳ more accurate representation of the derivative

Ex Estimate the 1st derivative of $f(x)$ at $x=0.5$ with a step size $h=0.5$ $h=0.25$

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.25$$

True value: $f'(x) = -0.4x^3 - 0.45x^2 - x - 0.25$

$$f'(0.5) = -0.925$$

Solution: For $h=0.5$

$x_{i-1} = 0$, $f(x_{i-1}) = 1.25 = f(0)$

$x_i = 0.5$, $f(x_i) = 0.925 = f(0.5)$

$x_{i+1} = 1$, $f(x_{i+1}) = 0.25 = f(1)$

Forward diff:

$$f'(0.5) \cong \frac{f(1) - f(0.5)}{1 - 0.5} = \frac{0.2 - 0.925}{0.5} = -1.45$$

$$|\varepsilon_t| = 58.9\%$$

Backward diff

$$f'(0.5) \cong \frac{f(0.5) - f(0)}{0.5 - 0} = \frac{0.925 - 1.2}{0.5} = -0.55$$

$$|\varepsilon_t| = 39.7\%$$

Centered diff

$$f'(0.5) \cong \frac{f(1) - f(0)}{1 - 0} = \frac{0.2 - 1.2}{1} = -1.0, \quad |\varepsilon_t| = 9.6\%$$

For $h=0.25$

$x_{i-1} = 0.25$, $f(x_{i-1}) = 1.103 = f(0.25)$

$x_i = 0.5$, $f(x_i) = 0.925 = f(0.5)$

$x_{i+1} = 0.75$, $f(x_{i+1}) = 0.636 = f(0.75)$

Forward Diff:

$$f'(0.5) \cong \frac{f(0.75)-f(0.5)}{0.75-0.5} = \frac{0.636-0.925}{0.25} = -1.155$$

$$|\varepsilon_t| = 26.5\%$$

Backward Diff

$$f'(0.5) \cong \frac{f(0.5)-f(0.25)}{0.5-0.25} = \frac{0.925-1.103}{0.25} = -0.714$$

$$|\varepsilon_t| = 21.7\%$$

Centered Diff.

$$f'(0.5) \cong \frac{f(0.75)-f(0.25)}{0.75-0.25} = \frac{0.636-1.103}{0.5} = -0.934$$

$$|\varepsilon_t| = 2.4\%$$

↙ more accurate than forward and backward differences. (reduced by a factor of 4)

- Halving the step size approximately halves the error. for forward and backward diff. formulas.

Finite Difference Approximations of Higher Derivatives

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \dots \quad (3)$$

write a forward Taylor series expansion for $f(x_{i+2})$ i.t.o $f(x_i)$:

$$f(x_{i+2}) = f(x_i) + f'(x_i)(2h) + \frac{f''(x_i)}{2!}(2h)^2 + \dots \quad (2)$$

Multiply (1) by 2 and subtract from (2):

which can be solved for:

$$f''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2} + O(h)$$

Second forward finite difference

Second backward finite divided difference:

$$f''(x_i) = \frac{f(x_i) - 2f(x_{i-1}) + f(x_{i-2})}{h^2} + O(h)$$

centered version:

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1})}{h^2} + O(h^2)$$

$$= \frac{\frac{f(x_{i+1}) - f(x_i)}{h} - \frac{f(x_i) - f(x_{i-1})}{h}}{h}$$

(Derivative of a derivate)

Error Propagation

Goal: to study how errors in numbers can propagate through mathematical functions.

function: $f(x)$

$\tilde{x}$: an approximation of x

Goal: assess the effect of the discrepancy between x and $\tilde{x}$ on the value of the function:

$$\Delta f(\tilde{x}) = |f(x) - f(\tilde{x})|$$

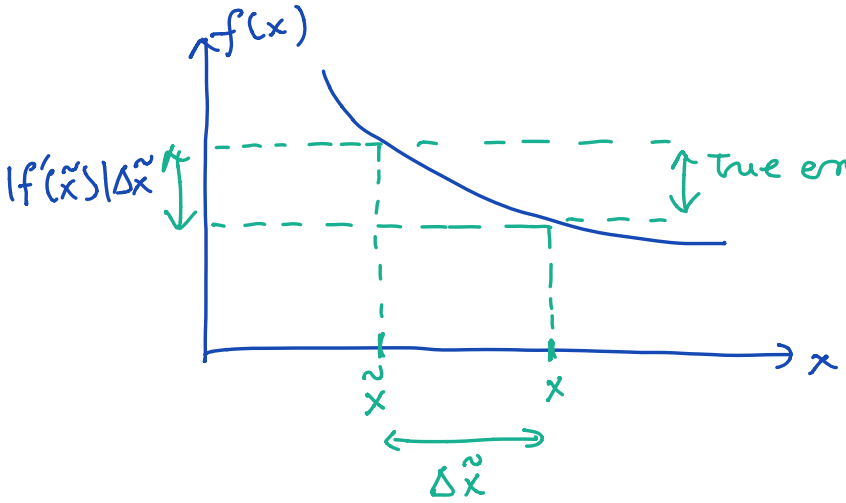
problem with evaluating $\Delta f(\tilde{x})$: x is unknown.

if $\tilde{x}$ is close to x and if $f(\tilde{x})$ is continuous and differentiable:

$$f(x) = f(\tilde{x}) + f'(\tilde{x})(x-\tilde{x}) + \underbrace{\frac{f''(\tilde{x})}{2!}(x-\tilde{x})^2}_{\text{Drop this}} + \cdots$$

$$f(x) - f(\tilde{x}) \cong f'(\tilde{x})(x-\tilde{x})$$

$$\underbrace{\Delta f(\tilde{x})}_{|f(x)-f(\tilde{x})|} = |f'(\tilde{x})|\,\Delta\tilde{x}$$



First order Error propagation

Ex Given $\tilde{x} = 2.5$ with an error of $\Delta\tilde{x} = 0.01$. Estimate the resulting error in the function $f(x) = x^3$

$$\Delta f(\tilde{x}) = |f'(\tilde{x})|\,\Delta\tilde{x}$$

$$\cong 3\tilde{x}^2\,\Delta\tilde{x}\Big|_{\tilde{x}=2.5} = 3(2.5)^2(0.01) = 0.1875$$

Since $f(2.5) = 15.625$

$$f(2.5) = 15.625 \pm 0.1875$$

or the true value lies between $\underbrace{15.4375}_{f(2.49)}$ and $\underbrace{15.8125}_{f(2.51)}$

Stability and Condition

- Condition of a mathematical problem is related to its sensitivity to changes in its input values.
- A computation is numerically unstable if the uncertainty of the input values is grossly magnified by the numerical method

- Use a first-order Taylor series:

$$f(x) = f(\tilde{x}) + f'(\tilde{x})(x - \tilde{x})$$

Relative error of $f(x)$:

$$\frac{f(x) - f(\tilde{x})}{f(x)} \simeq \frac{f'(\tilde{x})(x - \tilde{x})}{f(\tilde{x})}$$

The relative error of x is given by $\frac{x - \tilde{x}}{\tilde{x}}$

$$\text{Condition number} = \frac{\text{Relative error of } f(x)}{\text{Relative error of } x} = \frac{\tilde{x} f'(\tilde{x})}{f(\tilde{x})}$$

a measure of the extent to which an uncertainty in x is magnified by $f(x)$:

if cond.num $= 1 \Rightarrow$ function relative error is identical to the relative error of x.

if cond.num $> 1 \Rightarrow$ relative error is amplified.

" " very large, that function is called a ill-conditioned.

Ex Compute and interpret the condition number for

$f(x) = \tan x$ for $\tilde{x} = \frac{\pi}{2} + 0.1\left(\frac{\pi}{2}\right)$

$\tilde{x} = \frac{\pi}{2} + 0.01\left(\frac{\pi}{2}\right)$

$$\text{cond.num} = \frac{\tilde{x} f'(\tilde{x})}{f(\tilde{x})}$$

$$= \frac{\tilde{x}\,\frac{1}{\cos^2 \tilde{x}}}{\tan \tilde{x}}$$

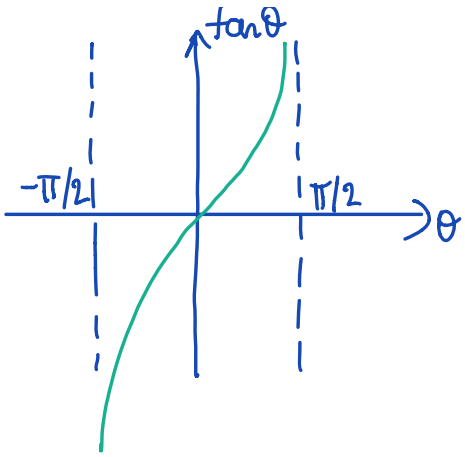
put $\tilde{x} = \frac{\pi}{2} + 0.1\left(\frac{\pi}{2}\right)$ $\longrightarrow$ cond.num $= -11.2$

$\Rightarrow$ Function is ill-conditioned

put $\tilde{x} = \frac{\pi}{2} + (0.01)\frac{\pi}{2}$ $\Rightarrow$ cond.num $= -101$

ill-conditioned

major source of ill-conditioning is the derivative



Around $\frac{\pi}{2}$ the tanget approache positive and negative infinity.